IE 7615 Discriminative Deep Learning Project

Milestone 2: Multi-Object Image Detection with YOLOv8s

Group 2

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1. Dataset and Problem Overview

Milestone 2 extends the single-object classification task in Milestone 1 to multi-object detection. Given one image that contains many objects, the detector must return the Object ID and the location (bounding box) of every object. We used all 73 final Object IDs (OBJ001–OBJ073) from the class object sheet and followed the instructors' Milestone 2 clarifications: multi-object images are grids of 224 × 224 photographs placed side by side, from 2 × 2 up to 5 × 5.

The shared collection contains 6,964 object photographs; the 379 background-only photographs (marked _bg_) are not used, because grid images have no background. Before splitting, the photographs were cleaned in code. OBJ054_017 was excluded because it is not a photo of the object. Exact duplicates were found by comparing the decoded pixels, and 16 extra copies were removed (in OBJ011, OBJ022, OBJ023, OBJ028, OBJ029, OBJ062 and OBJ066), keeping one copy of each. Every photograph was rotated according to its EXIF orientation and resized to 224 × 224, which fixes the two photographs with a different size (OBJ003_072, 224 × 192, and OBJ023_001, 225 × 224). This leaves 6,947 unique photographs.

The unique photographs were split 70 / 15 / 15 into training, validation and test sets, stratified by Object ID with seed 42 (the same strategy as Milestone 1). The split was made before any multi-object image was generated, and a check in the notebook confirms that no photograph, or a copy of it, appears in more than one split.

Source photographs	Train	Validation	Test	Total
Unique object photographs	4,862	1,042	1,043	6,947
Share	70%	15%	15%	100%

Table 1. Unique object photographs after cleaning, split before grid generation.

2. Multi-Object Image Generation and Labels

2.1 Bounding boxes for the single photographs

The shared dataset provides class labels (the folder names) but no bounding boxes. Each photograph shows one object, so it needs one box. We created the boxes automatically with two different open-vocabulary detectors, YOLO-World v2 (x) and Grounding DINO (tiny). Both were prompted with the object's OBJECT_NAME from the class sheet (for example "Pencil Case" for OBJ036) and returned up to three candidate boxes per photograph. As the instructors suggested, comparing the two models catches many mistakes that either model makes alone.

A box is accepted when the two models agree, meaning that their boxes overlap with IoU ≥ 0.5; the final box is the average of the two. When the models do not agree, a single model's box is accepted only if that model is confident (Grounding DINO score ≥ 0.35 or YOLO-World score ≥ 0.20). The remaining photographs are left out of the grid images, but never more than 20% of any Object ID: when more photographs were unreliable, the highest-scoring boxes were kept ("rescued") so that the limit was respected.

Label status	Rule	Photographs
Both models agree	Top boxes overlap with IoU ≥ 0.5	4,712 (67.8%)
Agree on 2nd/3rd candidate	A lower-ranked candidate matches (IoU ≥ 0.5)	568 (8.2%)
Grounding DINO only	No agreement, score ≥ 0.35	1,180 (17.0%)
YOLO-World only	No agreement, score ≥ 0.20	8 (0.1%)
Rescued	Kept to respect the 20% limit	130 (1.9%)
Left out	No reliable box from either model	349 (5.0%)
Total		6,947 (100%)

Table 2. Outcome of the automatic box labelling.

The two models agreed on 5,280 photographs (76.0%), and where they agreed their boxes overlapped with a mean IoU of 0.944. We checked the labels visually with three figures generated by the notebook: a random sample of 48 photographs from all Object IDs (Figure 1), 48 of the 1,667 photographs where the models did not agree, and all 95 photographs of our own object, OBJ036 (Figure 2). In the random sample, 38 of the 43 kept boxes are correct; the other 5 are on another item in the photograph or much too large (for example OBJ042_090, where both models boxed a game controller instead of the mouth guard case). For OBJ036, 69 of the 76 kept boxes are correct; the other 7 cover the rug, the chair or another item instead of the pencil case, mostly from the “agree on 2nd/3rd candidate” rule. Based on these checks, about one kept box in ten is wrong. These boxes were not corrected by hand before training (see Limitations).

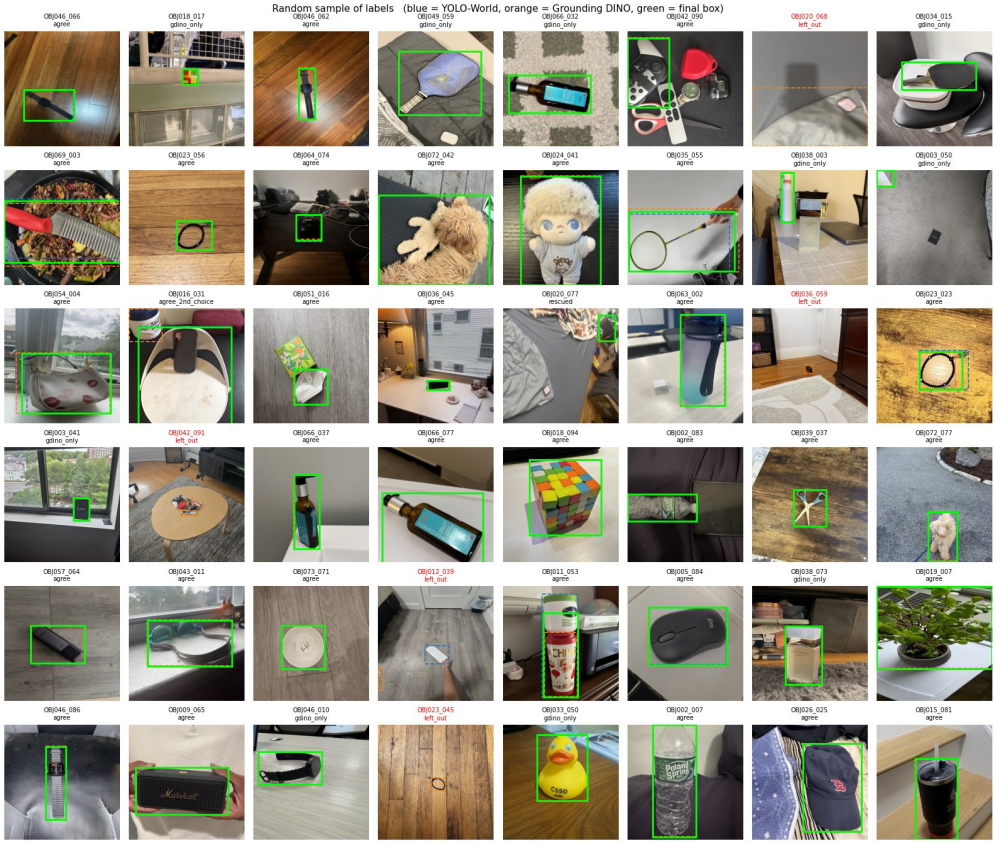


Figure 1. Random sample of 48 automatic labels (blue dashed = YOLO-World, orange dashed = Grounding DINO, green = final box; red title = left out).



Figure 2. All 95 OBJ036 photographs with their automatic labels, checked by hand.

In total, 349 photographs (5.0%) from 42 Object IDs were left out, in every case because neither model found the object with enough confidence. The check figures show that these are mostly photographs in which the object is very small or far away, dark on a dark background, or hidden among other items. Table 3 lists the number left out per Object ID. The most affected Object IDs, with 18–19 photographs left out each, are OBJ020, OBJ025, OBJ034, OBJ036, OBJ037, OBJ042, OBJ050, OBJ059 and OBJ068. No Object ID exceeded the 20% limit, and the other 31 Object IDs had no photograph left out.

Object ID	Left out	Object ID	Left out	Object ID	Left out	Object ID	Left out
OBJ003	8	OBJ026	1	OBJ041	12	OBJ059	19
OBJ004	6	OBJ029	1	OBJ042	18	OBJ060	11
OBJ007	1	OBJ030	6	OBJ044	4	OBJ061	12
OBJ009	1	OBJ031	2	OBJ046	2	OBJ062	2
OBJ012	8	OBJ032	11	OBJ049	6	OBJ064	1
OBJ013	9	OBJ033	8	OBJ050	19	OBJ065	1
OBJ014	11	OBJ034	19	OBJ051	11	OBJ068	19
OBJ017	1	OBJ036	19	OBJ052	1	OBJ069	1
OBJ020	19	OBJ037	19	OBJ054	14	OBJ070	1
OBJ023	7	OBJ038	2	OBJ057	2		
OBJ025	18	OBJ040	15	OBJ058	1		

Table 3. Photographs left out per Object ID (only the 42 Object IDs with at least one; reason: no reliable box).

2.2 Grid images

The multi-object images are built by placing the 224 × 224 photographs side by side in the five grid sizes of the instructors' table (Table 4). We generated 800 training, 150 validation and 150 test grids, with the five sizes in equal numbers in every split (Figure 3). Training grids use only training photographs, validation grids only validation photographs and test grids only test photographs. Within a split, photographs are drawn from a shuffled deck, so that each one is used about equally often and never twice in the same grid. Photographs that were left out in Section 2.1 are not used. Examples of test grids with their true boxes are shown in Figure 6.

For data augmentation, each tile of a training grid is randomly mirrored (with its box mirrored too) and receives random brightness, contrast and colour changes of up to ±25%; validation and test grids are not augmented. YOLOv8 adds its own augmentation during training (mosaic, scaling, translation, HSV changes and flips). Each object box is moved into grid coordinates and stored in YOLO format (class_id x_center y_center width height, normalised by the grid width and height). The data.yaml file defines the train, validation and test folders and the names of the 73 Object IDs.

Grid	Objects per image	Image size	Train	Validation	Test
2 × 2	4	448 × 448	160	30	30
3 × 3	9	672 × 672	160	30	30
4 × 4	16	896 × 896	160	30	30
5 × 4	20	1120 × 896	160	30	30
5 × 5	25	1120 × 1120	160	30	30
Total images			800	150	150
Total objects			11,840	2,220	2,220

Table 4. Generated grid dataset (1,100 multi-object images, 16,280 objects).

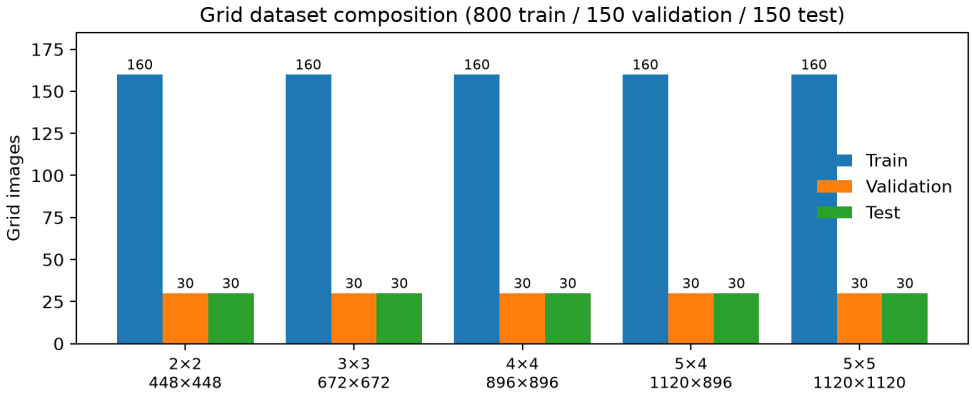


Figure 3. Number of grid images per grid size and split.

3. Model Training

A YOLOv8s detector initialised with COCO-pretrained weights (yolov8s.pt, 11.2 M parameters) was fine-tuned for the 73 Object IDs (transfer learning): the pretrained backbone and neck are reused, the detection head is replaced for 73 classes, and the whole network is trained on the grid images. Training was performed in Google Colab on an NVIDIA Tesla T4 GPU. Following the instructors' tip, the input size was set to 1120 px, close to the largest grid, so that objects in 5 × 5 grids are not shrunk; at the default 640 px, each 224 px cell would be reduced to about 128 px. All 50 epochs were completed (early stopping did not trigger), and the checkpoint with the best validation score was kept for testing.

Training setting	Value
Model	YOLOv8s (COCO-pretrained, 11.2 M parameters)
Epochs	50 (all completed)
Input resolution	1120 × 1120
Batch size	8
Optimizer	AdamW, learning rate 0.00013 (chosen automatically)
Early stopping patience	15
Augmentation	Tile flips and colour changes; YOLOv8 defaults (mosaic off for the last 10 epochs)
Random seed	42
Number of output classes	73
Hardware	Tesla T4 GPU
Training time	0.683 hours (~41 min)

Table 5. YOLOv8s training configuration.

Figure 4 shows the training losses and the validation metrics for every epoch. The classification loss falls from 4.39 in the first epoch to 0.29 in the last. Validation mAP@0.50 reaches 0.90 at epoch 20 and then stays around 0.91, so longer training would bring little gain. The selected checkpoint reaches Precision 0.912, Recall 0.872, mAP@0.50 0.908 and mAP@0.50:0.95 0.839 on the validation grids.

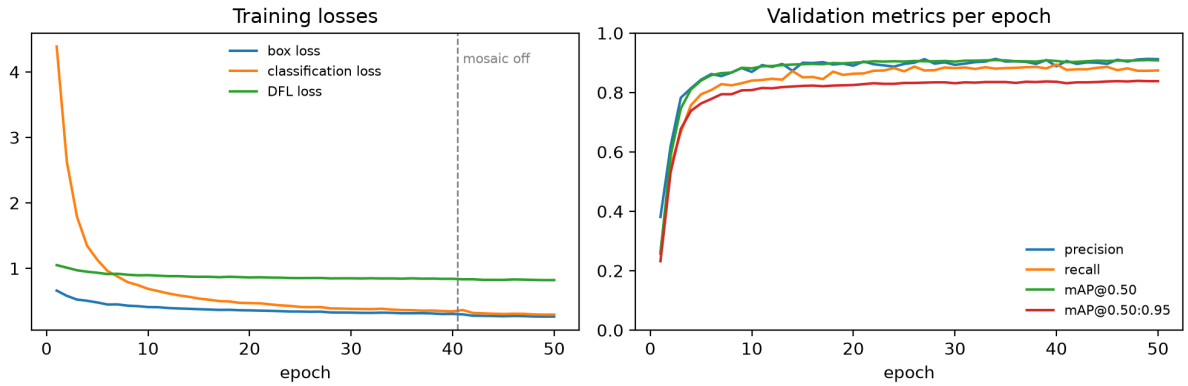


Figure 4. Training losses and validation metrics per epoch.

4. Experimental Results

We evaluated the selected checkpoint on the held-out test grids (150 images, 2,220 objects), which are built only from test photographs, using the notebook's split="test" evaluation call at 1120 px. Table 6 reports the values printed by the notebook; precision and recall are measured at the confidence threshold with the best F1 score.

Metric	Validation	Test
Precision	0.912	0.902
Recall	0.872	0.896
mAP@0.50	0.908	0.909
mAP@0.50:0.95	0.839	0.850

Table 6. YOLOv8s detection performance on the validation and test grids.

The test results match the validation results, so the model is not overfitted to the validation grids that were used to choose the checkpoint. Because each box now surrounds the object inside its photograph rather than the whole photograph, mAP@0.50:0.95 (0.850) is lower than mAP@0.50 (0.909): the model has to place each box accurately, not only find the grid cell.

Grid	Test images	Objects	Precision	Recall	mAP@0.50	mAP@0.50:0.95
2 × 2	30	120	0.842	0.841	0.929	0.858
3 × 3	30	270	0.885	0.848	0.926	0.880
4 × 4	30	480	0.867	0.894	0.919	0.874
5 × 4	30	600	0.891	0.865	0.898	0.844
5 × 5	30	750	0.901	0.893	0.916	0.855
All	150	2,220	0.902	0.896	0.909	0.850

Table 7. Test results per grid size.

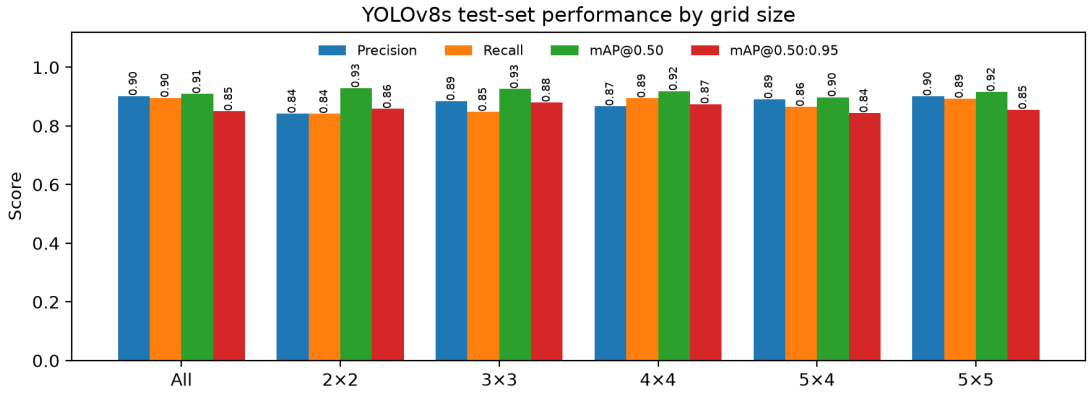


Figure 5. Test-set Precision, Recall, mAP@0.50 and mAP@0.50:0.95 for each grid size.

Performance is similar for all grid sizes: mAP@0.50 ranges from 0.898 (5 × 4) to 0.929 (2 × 2), so the model handles 25 objects per image almost as well as 4. Precision is lowest for the 2 × 2 grids (0.842), which contain only 120 test objects, so this value is the least certain.

To measure the system as it would be used, we also ran the detector on every test grid with a fixed confidence threshold of 0.5 and counted a detection as correct when the Object ID is right and the box overlaps the true box with IoU ≥ 0.5 (Table 8). At this threshold, 86.5% of the test objects are found with the correct ID and location, and 92.2% of the reported detections are correct. Only 22 of the 150 test grids (15%) are completely correct, because a single miss among up to 25 objects is enough to fail a grid.

Measure (all 150 test grids, confidence 0.5)	Value
Test objects	2,220
Correct detections (right ID, IoU ≥ 0.5)	1,920
Wrong ID or extra detections	162
Missed objects	300
Precision	0.922
Recall	0.865
Grids with every object correct and nothing extra	22 of 150

Table 8. Detection results on all test grids at a fixed confidence threshold of 0.5.

5. Detection Examples and Error Analysis

Figure 6 shows four test grids that contain our object, OBJ036 (3 × 3, 4 × 4, 5 × 4 and 5 × 5). OBJ036 is detected in all four, with confidence 0.99–1.00. Over the four grids, 61 of the 70 objects are detected with the correct ID and location, and the model makes 3 wrong detections.

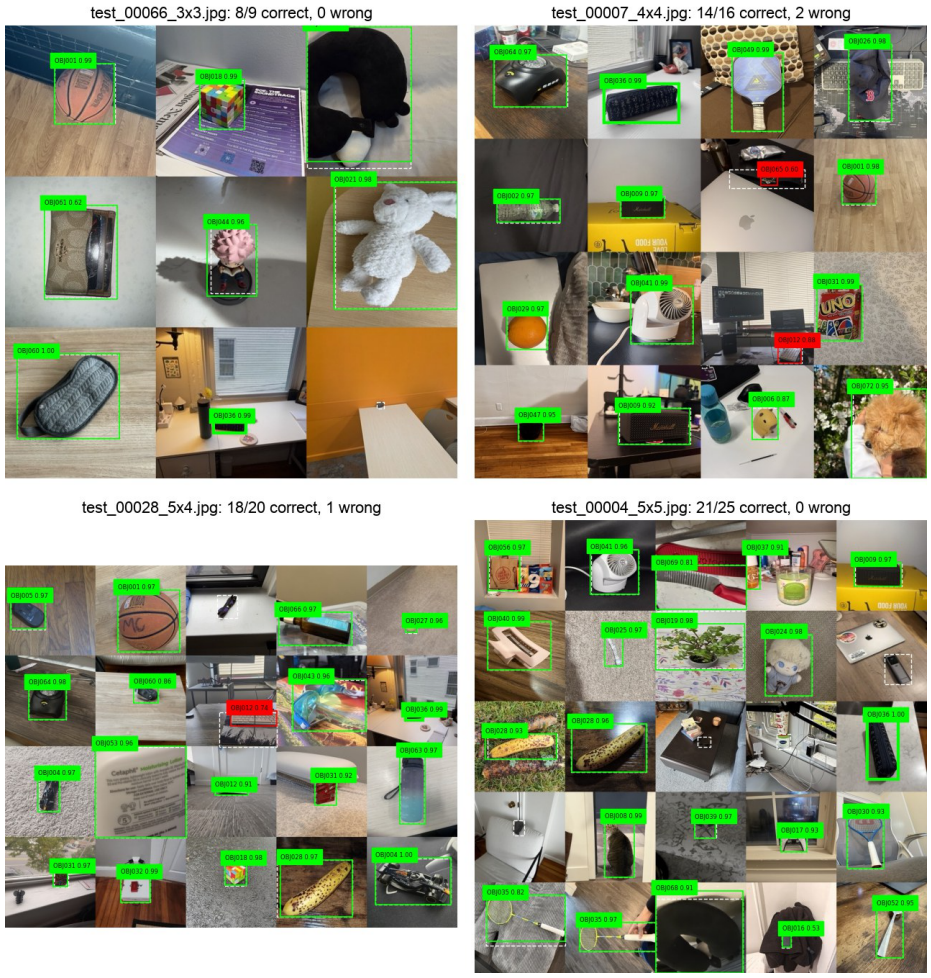


Figure 6. Example detections on test grids containing OBJ036 (white dashed = true box, green = correct, red = wrong; thick box = OBJ036).

Table 9 lists selected results for the 5 × 5 grid test_00004_5x5.jpg. The model reports 21 objects, all with the correct Object ID and with boxes that match the true boxes within a few pixels, including OBJ036 in row 3, column 5 (confidence 1.00). It misses four objects. Three of them (OBJ014, OBJ011 and OBJ060) are among the smallest in the image, with true boxes only 28–34 px wide inside a 224 px cell, because these photographs were taken from far away. OBJ016, which is about as small, was found but only with a confidence of 0.53.

Object ID	Row, column	Confidence	Predicted (x1, y1, x2, y2)	True (x1, y1, x2, y2)
OBJ036	3, 5	1.00	(971, 488, 1048, 640)	(972, 488, 1048, 639)
OBJ019	2, 3	0.98	(448, 255, 668, 370)	(450, 254, 663, 367)
OBJ016	5, 4	0.53	(761, 1017, 783, 1056)	(761, 1017, 783, 1057)
OBJ057	2, 5	missed	—	(1014, 330, 1081, 406)
OBJ014	3, 3	missed	—	(551, 536, 584, 560)
OBJ011	3, 4	missed	—	(748, 528, 782, 565)
OBJ060	4, 1	missed	—	(95, 743, 123, 779)

Table 9. Selected detections in test_00004_5x5.jpg (21 of 25 objects detected, no wrong detections).

Performance varies by Object ID. On the validation grids, the weakest Object IDs are OBJ050 (Hairbrush, mAP@0.50 0.26), OBJ061 (Card wallet, 0.56), OBJ010 (Nike sneakers, 0.60), OBJ051 (Tissue, 0.65) and OBJ058 (Footstool, 0.66). OBJ050 also had 19 photographs left out during labelling, and the object (a blue wide-tooth comb) looks similar to

OBJ059 (Comb), so both weak labels and similar objects are likely causes. Our own object, OBJ036, reaches precision 0.86, recall 0.90, mAP@0.50 0.941 and mAP@0.50:0.95 0.925 on the test grids.

6. Comparison and Analysis

6.1 Relationship to Milestone 1

Milestone 1 selected EfficientNetB0 for single-object classification (97.4% test accuracy). Milestone 2 uses YOLOv8s to both identify and locate many objects in one image. Classification accuracy and detection mAP are different measures and should not be compared directly. Both milestones use pretrained networks (transfer learning), the same 73 Object IDs and a stratified 70 / 15 / 15 split with seed 42; in Milestone 2, exact duplicates were removed before the split.

6.2 Comparison with our first composite pipeline

Our first version pasted 2–5 resized photographs onto a 640 × 640 background photograph and used each whole pasted photograph as the box. It reached mAP@0.50 0.962 and mAP@0.50:0.95 0.962 on its own test set, but the two metrics were identical because the boxes only had to match the edges of the pasted squares, not the objects. That version also did not use grids, did not remove duplicate photographs (13 groups of identical photographs fell into different splits) and did not exclude OBJ054_017. The final pipeline addresses all of these points. Its scores are lower (mAP@0.50 0.909, mAP@0.50:0.95 0.850) but more meaningful: it handles up to 25 objects per image and has to locate each object inside its cell. The two sets of results come from different test sets and are not directly comparable.

7. Final Model Selection and Justification

YOLOv8s fine-tuned on the grid images is our final detector. It returns the Object ID and bounding box of every object, covers all 73 Object IDs, reaches 0.909 mAP@0.50 and 0.850 mAP@0.50:0.95 on 150 test grids, and keeps a similar level for every grid size up to 5 × 5. It is small (22.7 MB) and fast, with about 23 ms of inference per 1120 px image on a T4 GPU. We did not train other YOLO sizes or architectures on the same data, so we do not claim that YOLOv8s is the best possible choice; a larger model such as YOLOv8m is the natural next comparison.

Limitations

- The boxes were created automatically. Our checks suggest that about one kept box in ten is on another item or much too large; the same errors exist in the test labels, so the reported metrics include some label noise.
- 349 photographs (5.0%) were left out, mostly the hardest ones (very small, dark or cluttered), so the grids slightly under-represent difficult photographs.
- Small objects are the main failure: objects photographed from far away occupy only about 20–35 px of a 224 px cell and are often missed.
- Grid cells do not overlap, so occlusion between objects is not tested, and all test images are synthetic grids built the same way as the training images.
- Each Object ID has only about 30 objects in the test grids, so per-ID results are uncertain.
- Only one model size (YOLOv8s) and one random split were evaluated.

Next Steps

Correct the wrong boxes found in the label checks (the notebook supports manual exclusions and manual boxes) and retrain; compare YOLOv8s with a larger model such as YOLOv8m; improve small-object detection with scale augmentation or a higher input resolution; and test the model on photographs that show several physical objects at once, as planned for the Week 3 demonstration.

GitHub and Reproducibility

Group repository: https://github.com/yvonneliuYL/IE7615-Group2-Discriminative-Project